

Tier 6: Group-Based Trajectory Modelling (GBTM)

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Overview

This report implements a genuine population-level GBTM, following the design of Yang et al. (2022, J Intensive Med). The literature review identified GBTM as a method used on MIMIC-IV sepsis cohorts but absent from this pipeline, which previously used a within-patient HMM to cluster states rather than trajectory shapes across patients. A latent-class growth model is fit on repeated 24-hour SOFA scores via flexmix, and class membership is then linked to mortality via Cox regression and to incident Sepsis-3 via logistic regression.

Load Libraries

We load flexmix for the latent-class growth model and survival for the Cox regression linking trajectory class to mortality.

```
suppressPackageStartupMessages({  
  library(flexmix); library(survival)  
})  
set.seed(486649)
```

Locate and Read Data

The helper function searches candidate paths for the required CSV files. We read the sampled long panel (2,500 stays from script 13) and the cohort-level data with demographics and outcomes.

```
find_data <- function(f) {  
  cands <- c(file.path(".././processed_data", f),  
             file.path("processed_data", f), f)  
  hit <- cands[file.exists(cands)][1]  
  if (is.na(hit)) stop("cannot find ", f); hit  
}  
OUT <- dirname(find_data("panel_sample_long.csv"))  
FIG <- if (dir.exists("figure")) "figure" else { dir.create("figure"); "figure" }  
  
panel <- read.csv(find_data("panel_sample_long.csv"))
```

```
cohort <- read.csv(find_data("cohort.csv"))[, c("stay_id", "age", "gender",
                                                "hospital_expire_flag", "los_hours")]
```

Build the First-24h SOFA Trajectory Sample

We restrict the panel to the first 24 hours and keep only stays with at least 12 of the 25 possible hourly rows, ensuring a stable polynomial fit. This drops a small number of stays with sparse early data.

```
# ---- Build the first-24h SOFA trajectory sample ----
# Keep stays with enough repeated points (>=12 of the 25 possible hourly rows)
# for a stable polynomial fit; this drops 62/2500 stays with sparse early data.
d <- panel[panel$hour <= 24, c("stay_id", "hour", "sofa_total")]
n_obs <- table(d$stay_id)
keep_ids <- names(n_obs[n_obs >= 12])
d <- d[as.character(d$stay_id) %in% keep_ids, ]
d$stay_id <- factor(d$stay_id)
cat(sprintf("GBTM sample: %d stays, %d stay-hours (first 24h, >=12 obs/stay)\n",
            length(unique(d$stay_id)), nrow(d)))
```

```
## GBTM sample: 2438 stays, 58267 stay-hours (first 24h, >=12 obs/stay)
```

Fit Latent-Class Growth Models

We fit latent-class growth models for $k = 2$ through 4 classes and select the best model by BIC. The `| stay_id` syntax in the formula tells flexmix to assign all rows of one stay to the same latent class, which is exactly the GBTM formulation. Multiple random restarts are used at each k to avoid local optima.

```
# ---- Fit latent-class growth models for k = 2..4, select by BIC ----
# `sofa_total ~ poly(hour, 2) | stay_id`: the `| stay_id` term is flexmix's
# grouped-mixture syntax -- it assigns ALL rows of one stay_id to the SAME
# latent class (a trajectory class), rather than mixing at the observation
# level. This is exactly the latent-class growth model (GBTM) formulation.
# stepFlexmix repeats each k with multiple random starts and keeps the best
# converged one; `minprior` prunes components that shrink to near-empty
# during EM (a handful of restarts otherwise degenerate to a zero-variance
# component and abort with a non-finite log-likelihood -- stepFlexmix just
# discards those restarts and keeps the ones that converge cleanly).
cat("fitting latent-class growth models (k=2..4; this takes a minute) ...\n")
```

```
## fitting latent-class growth models (k=2..4; this takes a minute) ...
```

```
sf <- stepFlexmix(sofa_total ~ poly(hour, 2) | stay_id, data = d, k = 2:4,
                  nrep = 3, control = list(iter.max = 200, minprior = 0.05),
                  verbose = FALSE)
```

```
## Error in FLXfit(model = model, concomitant = concomitant, control = control, :
```

```
## 136 Log-likelihood: NA
```

```
## Error in FLXfit(model = model, concomitant = concomitant, control = control, :
```

```
## 118 Log-likelihood: NA
```

```
cat("\nBIC by k:\n"); print(sf)
```

```
##
```

```
## BIC by k:
```

```
##
```

```
## Call:
```

```
## stepFlexmix(sofa_total ~ poly(hour, 2) | stay_id, data = d, control = list(iter.max = 200,
##      minprior = 0.05), k = 2:4, nrep = 3, verbose = FALSE)
##
##   iter converged k k0    logLik      AIC      BIC      ICL
## 2    24      TRUE 2  2 -115455.2 230928.3 231009.1 231030.3
## 3    29      TRUE 3  3 -101903.3 203834.6 203960.2 203996.7
## 4    30      TRUE 3  4 -101903.3 203834.6 203960.3 203996.9

best <- getModel(sf, which = "BIC")
K <- ncol(best@posterior$scaled)
cat(sprintf("Selected K = %d classes by BIC\n", K))

## Selected K = 3 classes by BIC
```

Per-Stay Class Assignment

Each stay is assigned to the trajectory class with the highest posterior probability. Classes are then reordered by mean SOFA severity so that class 1 always corresponds to the mildest trajectory, providing a stable and interpretable numbering across reruns.

```
# ---- Per-stay class assignment (posterior mode) ----
cls_per_row <- clusters(best)
row_id <- d$stay_id
cls_by_stay <- tapply(cls_per_row, row_id, function(x) x[1]) # constant within stay
post <- best@posterior$scaled
post_max <- tapply(apply(post, 1, max), row_id, function(x) x[1])

assign_tab <- data.frame(stay_id = as.integer(names(cls_by_stay)),
                        class = as.integer(cls_by_stay),
                        posterior_prob = round(as.numeric(post_max), 4))

# Order classes by mean SOFA severity (low -> high) so "class 1" = mildest,
# a stable, interpretable ordering across reruns (flexmix's own numbering is
# arbitrary / order-of-convergence).
sev_order <- order(tapply(d$sofa_total, cls_by_stay[as.character(d$stay_id)], mean))
remap <- setNames(seq_along(sev_order), sev_order)
assign_tab$class <- remap[as.character(assign_tab$class)]
write.csv(assign_tab, file.path(OUT, "tier6_gbtm_assignments.csv"), row.names = FALSE)
```

Mean Trajectory Per Class and Trajectory Plot

We compute the mean SOFA score at each hour for each trajectory class, then plot the resulting curves. This figure is the primary visual output of the GBTM analysis.

```
# ---- Mean trajectory per class (figure input) ----
d$class <- remap[as.character(cls_by_stay[as.character(d$stay_id)])]
means <- aggregate(sofa_total ~ hour + class, data = d, FUN = mean)
means <- means[order(means$class, means$hour), ]
write.csv(means, file.path(OUT, "tier6_gbtm_class_means.csv"), row.names = FALSE)

png(file.path(FIG, "gbtm_trajectories.png"), width = 1100, height = 850, res = 200)
cols <- c("steelblue", "darkorange", "firebrick", "purple")[1:K]
plot(NULL, xlim = c(0, 24), ylim = range(means$sofa_total),
     xlab = "ICU hour", ylab = "Mean SOFA total",
     main = sprintf("GBTM: %d latent SOFA trajectory classes (first 24h)", K))
for (k in 1:K) {
```

```

  m <- means[means$class == k, ]
  lines(m$hour, m$sofa_total, col = cols[k], lwd = 2, type = "b", pch = 19)
}
legend("topleft", paste("Class", 1:K, c("(mildest)", rep("", K - 2), "(most severe)")[1:K]),
      col = cols, lwd = 2, bty = "n", cex = 0.75)
dev.off()

## pdf
## 2

```

Link Trajectory Class to Mortality via Cox Regression

Following Yang et al. (2022), we fit a Cox proportional hazards model with trajectory class membership as the predictor and in-hospital mortality as the event. Time-to-event is hospital length of stay in hours.

```

# ---- Link trajectory class to mortality via Cox regression ----
# Following Yang et al. (2022): GBM trajectory-group membership -> Cox model
# for mortality. Time-to-event = hospital LOS (hours); event = in-hospital death.
surv_d <- merge(assign_tab, cohort, by = "stay_id")
surv_d$class <- factor(surv_d$class, levels = 1:K)
surv_d$los_hours <- pmax(surv_d$los_hours, 0.5) # avoid zero survival time
cox_fit <- coxph(Surv(los_hours, hospital_expire_flag) ~ class + age, data = surv_d)
cox_s <- summary(cox_fit)
cox_tab <- data.frame(term = rownames(cox_s$coefficients),
                     HR = round(exp(cox_s$coefficients[, "coef"]), 3),
                     p_value = signif(cox_s$coefficients[, "Pr(>|z|)"], 3))
cat("\nCox model: trajectory class vs in-hospital mortality (ref = mildest class):\n")

##
## Cox model: trajectory class vs in-hospital mortality (ref = mildest class):
print(cox_tab, row.names = FALSE)

##      term      HR  p_value
## class2 2.525 2.22e-03
## class3 5.019 3.88e-08
##      age 1.029 1.74e-09

write.csv(cox_tab, file.path(OUT, "tier6_mortality_cox.csv"), row.names = FALSE)

```

Link Trajectory Class to Incident Sepsis-3

As a secondary analysis specific to this project, we test whether the SOFA trajectory shape over the first 24 hours is associated with ever meeting Sepsis-3 criteria, beyond the endpoint severity alone. This uses a logistic regression with trajectory class as the predictor.

```

# ---- Secondary link: trajectory class vs incident Sepsis-3 ----
# This project's own outcome of interest (not in Yang et al.): does the SOFA
# trajectory SHAPE over the first 24h associate with ever meeting Sepsis-3,
# beyond the trajectory's endpoint severity alone?
onset_d <- panel[!duplicated(panel$stay_id), c("stay_id", "sepsis3")]
onset_d <- merge(assign_tab, onset_d, by = "stay_id")
onset_d$class <- factor(onset_d$class, levels = 1:K)
onset_d$sepsis3 <- as.integer(as.logical(onset_d$sepsis3))
glm_fit <- glm(sepsis3 ~ class, data = onset_d, family = binomial)
glm_s <- summary(glm_fit)$coefficients

```

```

onset_tab <- data.frame(term = rownames(glm_s),
                        odds_ratio = round(exp(glm_s[, "Estimate"]), 3),
                        p_value = signif(glm_s[, "Pr(>|z|)"], 3))
cat("\nLogistic model: trajectory class vs incident Sepsis-3 (ref = mildest class):\n")

##
## Logistic model: trajectory class vs incident Sepsis-3 (ref = mildest class):
print(onset_tab, row.names = FALSE)

##      term odds_ratio p_value
## (Intercept)    0.077 8.48e-78
##      class2     6.862 9.84e-37
##      class3    18.809 1.46e-74

write.csv(onset_tab, file.path(OUT, "tier6_onset_assoc.csv"), row.names = FALSE)

cat("\nTier 6 (GBTM) complete.\n")

##
## Tier 6 (GBTM) complete.

```